

*Generate the variables required

```
gen real_minwage = minwage / cpi
```

```
gen ln_real_minwage = ln(real_minwage)
```

```
tab state
```

```
tab minwage
```

```
tab cpi
```

*Descriptive statistics

```
sum employed workpt workft minwage real_minwage ln_real_minwage
```

```
sum employed workpt workft minwage real_minwage ln_real_minwage, detail
```

*Graphs

```
histogram employed, percent normal title("Teen Employment Rate Distribution")
```

```
histogram ln_real_minwage, percent normal title("Log Real Min Wage Distribution")
```

```
preserve
```

```
collapse (mean) employed workpt workft minwage real_minwage, by(year)
```

* List summary

```
list year employed workpt workft minwage real_minwage
```

* Line graph: teen employment over time

```
twoway line employed year, title("Teen Employment Over Time") ytitle("Fraction  
Employed") xtitle("Year")
```

* Line graph: real min wage over time

```
twoway line real_minwage year, title("Real Minimum Wage Over Time") ytitle("Real  
Minimum Wage") xtitle("Year")
```

* Export table

```
export delimited using yearly_summary.csv, replace
```

```
restore
```

*Now we make the federal minimum wage column. Note: we use the Federal minimum wage value in March of that year as the CPS survey data is from March of every year 1990-2013. For certain years the new federal minimum wage is implemented in Apr(1990), Apr(1991), Oct(1996), Sep(1997), Jul(2007), Jul(2008), Jul(2009). Therefore, we take the previous year's Federal minimum wage for all these years.

```
gen fed_minwage = .
```

* As of March 1990, the federal minimum wage was still \$3.35

```
replace fed_minwage = 3.35 if year == 1990
```

* As of March 1991, the federal minimum wage was \$3.80 (Apr 1990 change)

```
replace fed_minwage = 3.80 if year == 1991
```

* As of March 1992–1996, it was \$4.25 (Apr 1991 change)

```
replace fed_minwage = 4.25 if inrange(year, 1992, 1996)
```

* As of March 1997, it was \$4.75 (Oct 1996 change)

```
replace fed_minwage = 4.75 if year == 1997
```

* As of March 1998–2007, it was \$5.15 (Sep 1997 change)

```
replace fed_minwage = 5.15 if inrange(year, 1998, 2007)
```

* As of March 2008, it was \$5.85 (Jul 2007 change)

```
replace fed_minwage = 5.85 if year == 2008
```

* As of March 2009, it was \$6.55 (Jul 2008 change)

```
replace fed_minwage = 6.55 if year == 2009
```

* As of March 2010 and onward, it was \$7.25 (Jul 2009 change)

```
replace fed_minwage = 7.25 if year >= 2010
```

* 1. Above Federal Min Wage?

```
gen state_above_fed = (minwage - fed_minwage > 0.009)
```

```
gen minwage_difference = minwage - fed_minwage
```

```
replace minwage_difference = 0 if minwage_difference < 0
```

* 2. IDENTIFY FIRST YEAR EACH STATE WAS ABOVE FED MIN WAGE

```
sort state year
```

```
gen year_dummy = state_above_fed
```

```
bysort state (year): gen year_difference = year if year_dummy == 1 & year_dummy[_n-1] != 1
```

* 3. CREATE A TREATMENT YEAR VARIABLE

```
gen treat_year = .
```

bysort state (year): replace treat_year = year if state_above_fed == 1 &
missing(treat_year)

bysort state (year): replace treat_year = treat_year[_n-1] if missing(treat_year)

* 5. DISPLAY FINAL CLEAN TABLE

list state treat_year, clean

restore

* Dropping states that had a minimum wage above the federal level in 1990.

* Since our dataset begins in 1990, we cannot observe a true pre-treatment period for these states, and including them would violate the parallel trends assumption required for Difference-in-Differences analysis.

drop if state == 2

drop if state == 6

drop if state == 9

drop if state == 15

drop if state == 19

drop if state == 23

drop if state == 25

drop if state == 27

drop if state == 33

drop if state == 38

drop if state == 41

drop if state == 42

drop if state == 44

drop if state == 50

drop if state == 53

drop if state == 55

*The treatment group consists of U.S. states that first implemented a minimum wage higher than the federal minimum wage after March 1990. These states did not have elevated minimum wages at the beginning of the dataset, allowing for a clean pre-treatment period. Treatment years vary by state depending on when the policy was first

adopted. This setup ensures compliance with the parallel trends assumption required for Difference-in-Differences analysis.

gen post = 0 if state == 4

replace post = 1 if state == 4 & year >= 2007

gen treatment = 1 if state == 4

* treatment for 2nd state

replace post = 0 if state == 5

replace post = 1 if state == 5 & year >= 2007

replace treatment = 1 if state == 5

* treatment for 3rd state

replace post = 0 if state == 8

replace post = 1 if state == 8 & year >= 2007

replace treatment = 1 if state == 8

* treatment for 4th state

replace post = 0 if state == 10

replace post = 1 if state == 10 & year >= 1997

replace treatment = 1 if state == 10

* treatment for 5th state

replace post = 0 if state == 11

replace post = 1 if state == 11 & year >= 1994

replace treatment = 1 if state == 11

* treatment for 6th state

replace post = 0 if state == 12

replace post = 1 if state == 12 & year >= 2006

replace treatment = 1 if state == 12

* treatment for 7th state

replace post = 0 if state == 17

replace post = 1 if state == 17 & year >= 2004

replace treatment = 1 if state == 17

* treatment for 8th state

replace post = 0 if state == 24

replace post = 1 if state == 24 & year >= 2007

replace treatment = 1 if state == 24

* treatment for 9th state

replace post = 0 if state == 26

replace post = 1 if state == 26 & year >= 2007

replace treatment = 1 if state == 26

* treatment for 10th state

replace post = 0 if state == 29

replace post = 1 if state == 29 & year >= 2007

replace treatment = 1 if state == 29

* treatment for 11th state

replace post = 0 if state == 30

replace post = 1 if state == 30 & year >= 2007

replace treatment = 1 if state == 30

* treatment for 12th state

replace post = 0 if state == 32

replace post = 1 if state == 32 & year >= 2007

replace treatment = 1 if state == 32

* treatment for 13th state

replace post = 0 if state == 34

replace post = 1 if state == 34 & year >= 1993

replace treatment = 1 if state == 34

* treatment for 14th state

replace post = 0 if state == 35

replace post = 1 if state == 35 & year >= 2008

replace treatment = 1 if state == 35

* treatment for 15th state

replace post = 0 if state == 36

replace post = 1 if state == 36 & year >= 2005

replace treatment = 1 if state == 36

* treatment for 16th state

replace post = 0 if state == 37

replace post = 1 if state == 37 & year >= 2007

replace treatment = 1 if state == 37

* treatment for 17th state

replace post = 0 if state == 39

replace post = 1 if state == 39 & year >= 2007

replace treatment = 1 if state == 39

* treatment for 18th state

replace post = 0 if state == 54

replace post = 1 if state == 54 & year >= 2007

replace treatment = 1 if state == 54

* These states never had a state minimum wage that exceeded the federal minimum wage during the sample period.

* They serve as valid controls under the assumption of parallel trends in the absence of treatment.

* 1st control

replace post = 0 if state == 1

replace treatment = 0 if state == 1

* 2nd control

replace post = 0 if state == 13

replace treatment = 0 if state == 13

* 3rd control

replace post = 0 if state == 16

replace treatment = 0 if state == 16

* 4th control

replace post = 0 if state == 18

replace treatment = 0 if state == 18

* 5th control

replace post = 0 if state == 20

replace treatment = 0 if state == 20

* 6th control

replace post = 0 if state == 21

replace treatment = 0 if state == 21

* 7th control

replace post = 0 if state == 22

replace treatment = 0 if state == 22

* 8th control

replace post = 0 if state == 28

replace treatment = 0 if state == 28

* 9th control

replace post = 0 if state == 31

replace treatment = 0 if state == 31

* 10th control

replace post = 0 if state == 40

replace treatment = 0 if state == 40

* 11th control

replace post = 0 if state == 45

replace treatment = 0 if state == 45

* 12th control

replace post = 0 if state == 46

replace treatment = 0 if state == 46

* 13th control

replace post = 0 if state == 47

replace treatment = 0 if state == 47

* 14th control

replace post = 0 if state == 48

replace treatment = 0 if state == 48

* 15th control

replace post = 0 if state == 49

replace treatment = 0 if state == 49

* 16th control

replace post = 0 if state == 51

replace treatment = 0 if state == 51

* 17th control

replace post = 0 if state == 56

replace treatment = 0 if state == 56

* Create interaction term for Difference-in-Differences (DiD)

* This captures the treatment effect by identifying observations

* in treated states (treatment == 1) and post-treatment years (post == 1).

* The variable 'treat_post' equals 1 only for treated states

* in the years after their minimum wage increase.

* This is the key variable used to estimate the Average Treatment

* Effect on the Treated (ATT) in a DiD regression framework.

```
gen treatment_post = treatment*post
```

```
xtset state year
```

```
gen employed_percent = employed*100
```

```
xtreg employed_percent treatment_post i.state i.year, fe
```

```
xtreg employed treatment_post i.year, fe
```

```
xtreg employed treatment_post i.year, re
```

```

gen employedf_percent = employedf *100
gen employedm_percent = employedm *100
xtreg employedf_percent treatment_post i.year, fe
xtreg employedm_percent treatment_post i.year, fe
gen work_fulltime_percent = workft *100
xtreg work_fulltime_percent treatment_post i.year, fe
gen work_parttime_percent = workpt *100
xtreg work_parttime_percent treatment_post i.year, fe
gen work_parttime_female_percent = workptf *100
gen work_parttime_male_percent = workptm *100
xtreg work_parttime_female_percent treatment_post i.year, fe
xtreg work_parttime_male_percent treatment_post i.year, fe
gen work_fulltime_female_percent = workftf *100
gen work_fulltime_male_percent = workftm *100
xtreg work_fulltime_female_percent treatment_post i.year, fe
xtreg work_fulltime_male_percent treatment_post i.year, fe
xtreg employed_percent treatment_post i.state i.year, fe

```

```

gen employed_percent = employed * 100

```

```

preserve

```

* Step 2: Collapse data to average employment by year & treatment group

```

collapse (mean) employed_percent, by(year treatment)

```

* Step 3: Plot DiD-style line graph

```

twoway (line employed_percent year if treatment == 1, lcolor(black) lpattern(solid)
lwidth(medthick) legend(label(1 "Treatment"))) (line employed_percent year if treatment
== 0, lcolor(black) lpattern(dash) lwidth(medthick) legend(label(2 "Control"))),
title("Teen Employment Rate: Treated vs Control States") ytitle("Employment Rate (%)")
xtitle("Year") legend(position(6) ring(0) row(1)) graphregion(color(white)) bgcolor(white)
restore

```

*Real Minimum wage

*Use clean dataset

* Generate nominal federal minimum wage by year

```
gen fed_minwage = .
```

```
replace fed_minwage = 3.35 if year == 1990
```

```
replace fed_minwage = 3.80 if year == 1991
```

```
replace fed_minwage = 4.25 if inrange(year, 1992, 1996)
```

```
replace fed_minwage = 4.75 if year == 1997
```

```
replace fed_minwage = 5.15 if inrange(year, 1998, 2007)
```

```
replace fed_minwage = 5.85 if year == 2008
```

```
replace fed_minwage = 6.55 if year == 2009
```

```
replace fed_minwage = 7.25 if year >= 2010
```

* Generate real minimum wages

```
gen real_minwage = minwage / cpi
```

```
gen real_fed_minwage = fed_minwage / cpi
```

* floating-point comparison to define treatment status

```
gen real_diff = real_minwage - real_fed_minwage
```

```
gen treated_year = (real_diff > 0.005)
```

* Identify first year each state is treated (i.e., real min > real fed min)

```
gen treat_year = .
```

```
sort state year
```

```
bysort state (year): replace treat_year = year if treated_year == 1 & treated_year[_n-1] != 1
```

```
bysort state (year): replace treat_year = treat_year[_n-1] if missing(treat_year)
```

```
list state year real_minwage real_fed_minwage treated_year treat_year if treated_year == 1, sepby(state)
```

```
drop if state == 6
```

```
drop if state == 41
```

```
drop if state == 44
```

```
drop if state == 53
```

```
drop if state == 9
```

```
* 1st treated state
```

```
gen post = 0 if state == 2
```

```
replace post = 1 if state == 2 & year >= 2003
```

```
gen treatment = 1 if state == 2
```

```
* 2nd treated state
```

```
replace post = 0 if state == 4
```

```
replace post = 1 if state == 4 & year >= 2007
```

```
replace treatment = 1 if state == 4
```

```
* 3rd treated state
```

```
replace post = 0 if state == 5
```

```
replace post = 1 if state == 5 & year >= 2007
```

```
replace treatment = 1 if state == 5
```

```
* 4th treated state
```

```
replace post = 0 if state == 8
```

```
replace post = 1 if state == 8 & year >= 2007
```

```
replace treatment = 1 if state == 8
```

```
* 6th treated state
```

```
replace post = 0 if state == 10
```

```
replace post = 1 if state == 10 & year >= 2001
```

replace treatment = 1 if state == 10

* 7th treated state

replace post = 0 if state == 11

replace post = 1 if state == 11 & year >= 1995

replace treatment = 1 if state == 11

* 8th treated state

replace post = 0 if state == 12

replace post = 1 if state == 12 & year >= 2007

replace treatment = 1 if state == 12

* 9th treated state

replace post = 0 if state == 15

replace post = 1 if state == 15 & year >= 1993

replace treatment = 1 if state == 15

* 10th treated state

replace post = 0 if state == 17

replace post = 1 if state == 17 & year >= 2005

replace treatment = 1 if state == 17

* 11th treated state

replace post = 0 if state == 19

replace post = 1 if state == 19 & year >= 2008

replace treatment = 1 if state == 19

* 12th treated state

replace post = 0 if state == 23

replace post = 1 if state == 23 & year >= 2003

replace treatment = 1 if state == 23

* 13th treated state

replace post = 0 if state == 25

replace post = 1 if state == 25 & year >= 2001

replace treatment = 1 if state == 25

* 14th treated state

replace post = 0 if state == 26

replace post = 1 if state == 26 & year >= 2007

replace treatment = 1 if state == 26

* 15th treated state

replace post = 0 if state == 27

replace post = 1 if state == 27 & year >= 2006

replace treatment = 1 if state == 27

* 16th treated state

replace post = 0 if state == 29

replace post = 1 if state == 29 & year >= 2007

replace treatment = 1 if state == 29

* 17th treated state

replace post = 0 if state == 34

replace post = 1 if state == 34 & year >= 1993

replace treatment = 1 if state == 34

* 18th treated state

replace post = 0 if state == 36

replace post = 1 if state == 36 & year >= 2006

replace treatment = 1 if state == 36

* 19th treated state

replace post = 0 if state == 39

replace post = 1 if state == 39 & year >= 2007

replace treatment = 1 if state == 39

* 20th treated state

replace post = 0 if state == 42

replace post = 1 if state == 42 & year >= 2007

replace treatment = 1 if state == 42

* 21st treated state

replace post = 0 if state == 50

replace post = 1 if state == 50 & year >= 2001

replace treatment = 1 if state == 50

* 22nd treated state

replace post = 0 if state == 55

replace post = 1 if state == 55 & year >= 2007

replace treatment = 1 if state == 55

* 1st control

replace post = 0 if state == 1

replace treatment = 0 if state == 1

* 2nd control

replace post = 0 if state == 13

replace treatment = 0 if state == 13

* 3rd control

replace post = 0 if state == 16

replace treatment = 0 if state == 16

* 4th control

replace post = 0 if state == 18

replace treatment = 0 if state == 18

* 5th control

replace post = 0 if state == 20

replace treatment = 0 if state == 20

* 6th control

replace post = 0 if state == 21

replace treatment = 0 if state == 21

* 7th control

replace post = 0 if state == 22

replace treatment = 0 if state == 22

* 8th control

replace post = 0 if state == 28

replace treatment = 0 if state == 28

* 9th control

replace post = 0 if state == 31

replace treatment = 0 if state == 31

* 10th control

replace post = 0 if state == 40

replace treatment = 0 if state == 40

* 11th control

replace post = 0 if state == 45

replace treatment = 0 if state == 45

* 12th control

replace post = 0 if state == 46

replace treatment = 0 if state == 46

* 13th control

replace post = 0 if state == 47

replace treatment = 0 if state == 47

* 14th control

replace post = 0 if state == 48

replace treatment = 0 if state == 48

* 15th control

replace post = 0 if state == 49

replace treatment = 0 if state == 49

* 16th control

replace post = 0 if state == 51

replace treatment = 0 if state == 51

* 17th control

replace post = 0 if state == 56

replace treatment = 0 if state == 56

*18th control

replace post = 0 if state == 33

replace treatment = 0 if state == 33

*19th control

replace post = 0 if state == 38

replace treatment = 0 if state == 38

*20th control

replace post = 0 if state == 24

replace treatment = 0 if state == 24

*21 control

replace post = 0 if state == 30

replace treatment = 0 if state == 30

*22 control

replace post = 0 if state == 32

replace treatment = 0 if state == 32

*23 control

replace post = 0 if state == 35

replace treatment = 0 if state == 35

*24 control

replace post = 0 if state == 37

replace treatment = 0 if state == 37

*25 control

replace post = 0 if state == 54

replace treatment = 0 if state == 54

```

gen treatment_post = treatment*post
gen employed_percent = employed*100
xtset state year
xtreg employed treatment_post i.year, fe
xtset state year
xtreg employed treatment_post i.year, fe
gen employedf_percent = employedf *100
gen employedm_percent = employedm *100
xtreg employedf_percent treatment_post i.year, fe
xtreg employedm_percent treatment_post i.year, fe
gen work_fulltime_percent = workft *100
xtreg work_fulltime_percent treatment_post i.year, fe
gen work_parttime_percent = workpt *100
xtreg work_parttime_percent treatment_post i.year, fe
gen work_parttime_female_percent = workptf *100
gen work_parttime_male_percent = workptm *100
xtreg work_parttime_female_percent treatment_post i.year, fe
xtreg work_parttime_male_percent treatment_post i.year, fe
gen work_fulltime_female_percent = workftf *100
gen work_fulltime_male_percent = workftm *100
xtreg work_fulltime_female_percent treatment_post i.year, fe
xtreg work_fulltime_male_percent treatment_post i.year, fe
xtreg employed_percent treatment_post i.state i.year, fe
preserve

```

*placebo test for both groups

```
gen placebo_treatment = treatment
```

```
gen placebo_post = (year >= 1993)
```

```
gen placebo_treat_post = placebo_treatment * placebo_post
```

```
reg employed_percent placebo_treatment placebo_post placebo_treat_post i.year,  
robust
```